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HI-MAPF

Towards Resource-Efficient Multi-Agent Deployment

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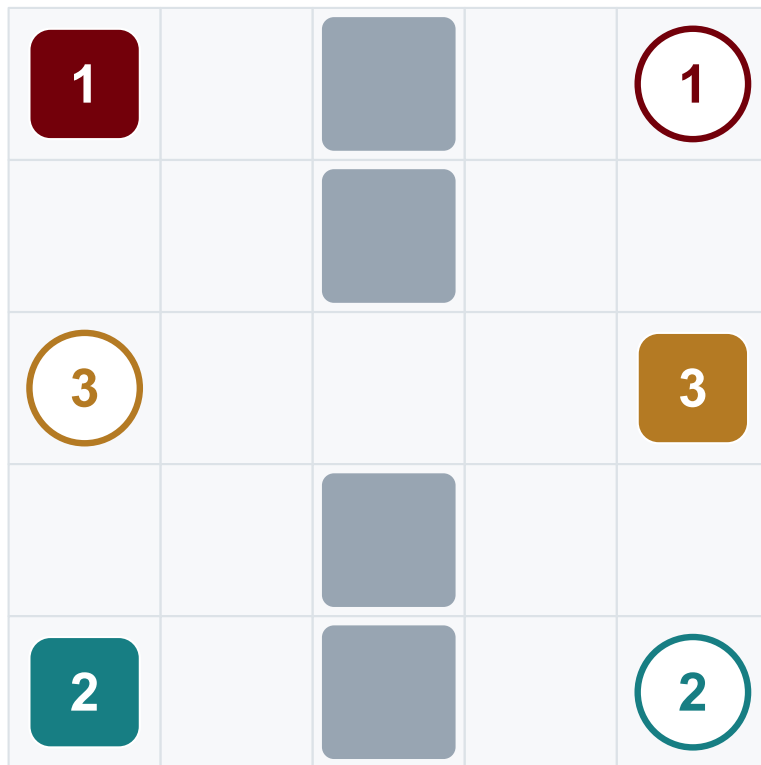
FROM PLANS TO REAL ROBOTS



Hybrid planning · event-triggered information

MAPF with resource-constrained robots

Find collision-free paths from every start to its assigned goal.



■ Start ○ Goal 3 agents · 5 × 5 grid

Coordination requires resources

Sensing

Local observations of the environment and other agents

Communication

Exchange of state, observations, and planned motion

Compute

Search, neural inference, and repeated replanning

Deployment cost needs its own measure.

I-MAPF: the information available to each agent

A unified formulation of what agent i knows at time t .

$$\mathcal{I}_i(t) = \left(\mathcal{M}_i, \mathcal{S}_i, \mathcal{O}_j, \mathcal{E}_{ij}(t) \right)$$

 $\mathcal{M}_i$

Environment knowledge

Map: vertices, edges, and obstacles

 $\mathcal{O}_j$

Other-agent information

States, goals, and path segments sensed or shared

 $\mathcal{S}_i$

Self information

Own state, goal, and planned path segment

 $\mathcal{E}_{ij}(t)$

Event-triggered signals

Collision / deadlock alerts and constraints

INFORMATION UNIT (IU)

$$1 \text{ IU} = (v, s, t, e) \in V \times S \times T \times E$$

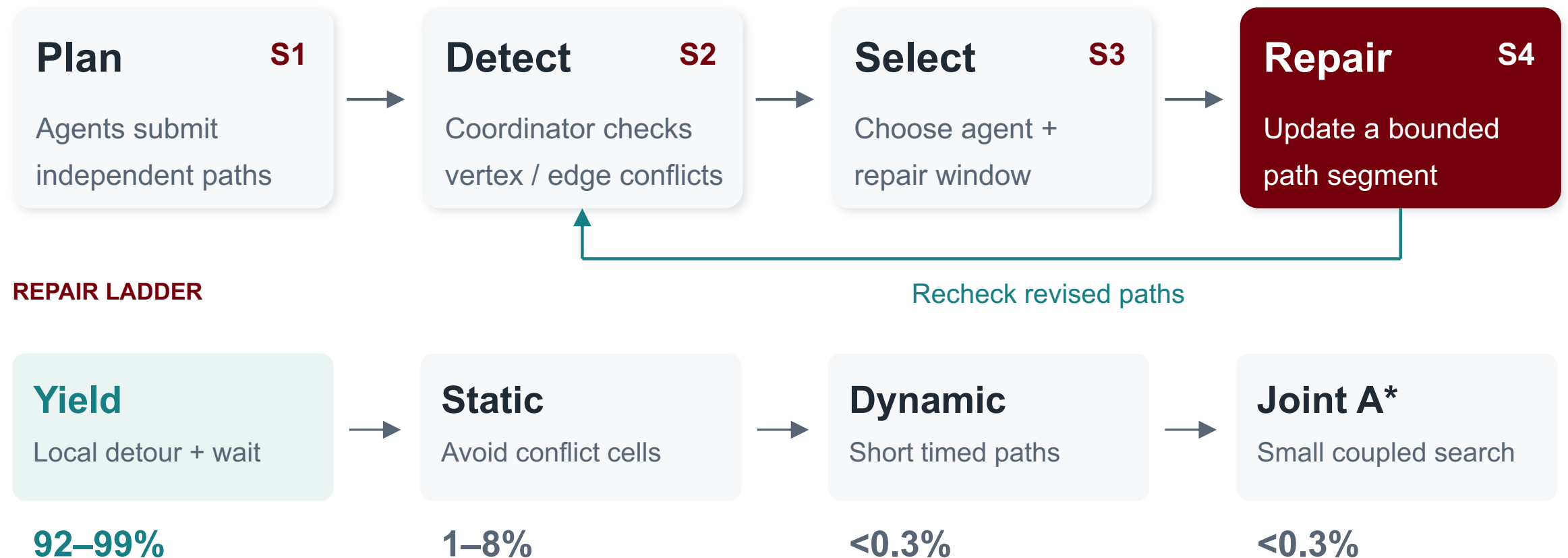
v = grid cell · s = cell state

t = optional timestep · e = optional message scalar

Sensing outputs and transmitted messages are finite multisets of IUs.

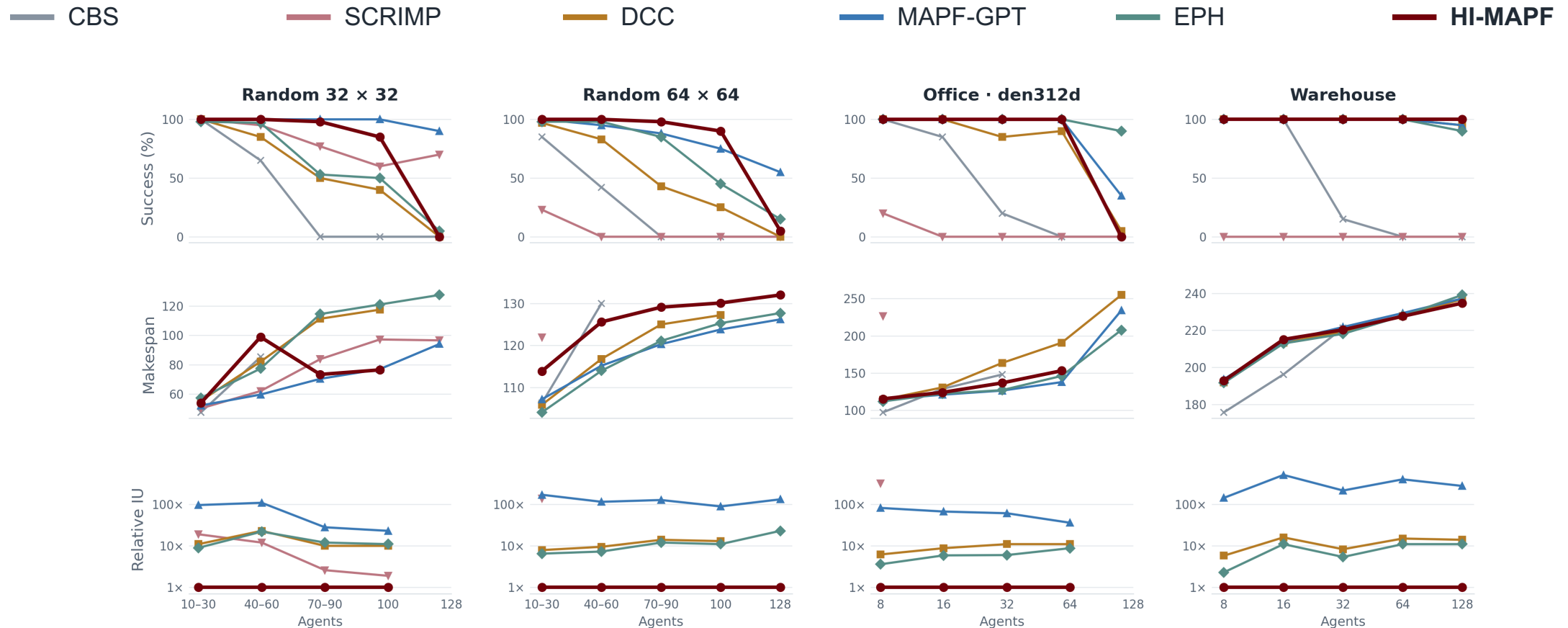
Plan independently. Repair selectively.

A lightweight coordinator intervenes when submitted paths conflict.



Less information across four benchmarks.

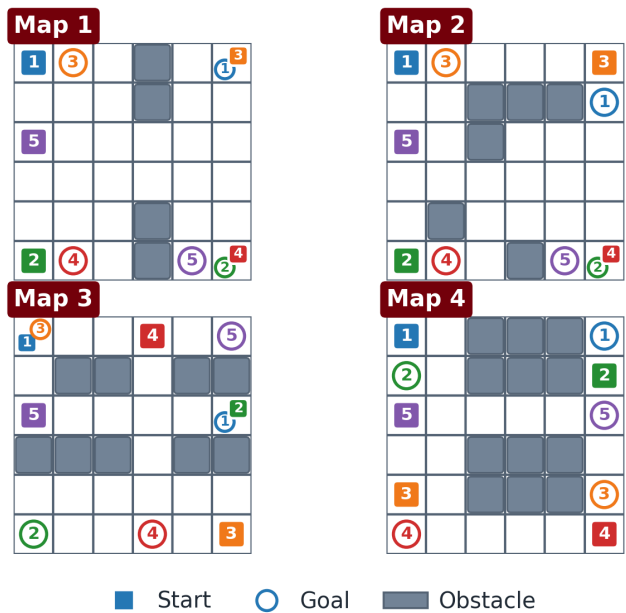
8–128 agents · Success, makespan and relative IU across all reported groups.



Lower planning cost on a real robot team.

Five TurtleBot4 robots · four 6 × 6 maps · five problem instances

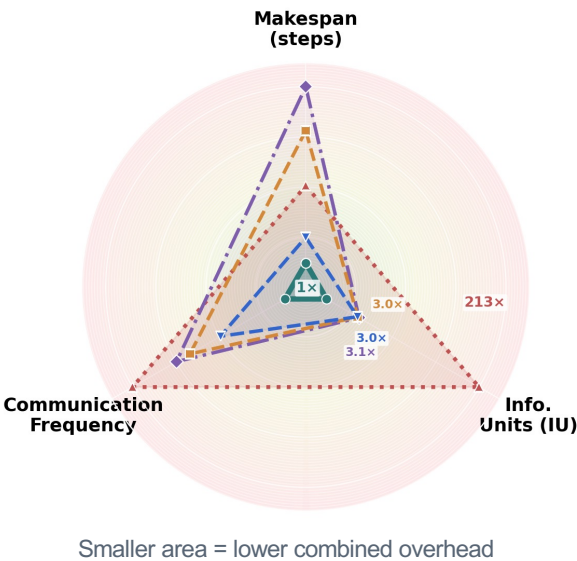
DEPLOYMENT MAPS



199 IU

3× below closest baselines · 213× below SCRIMP

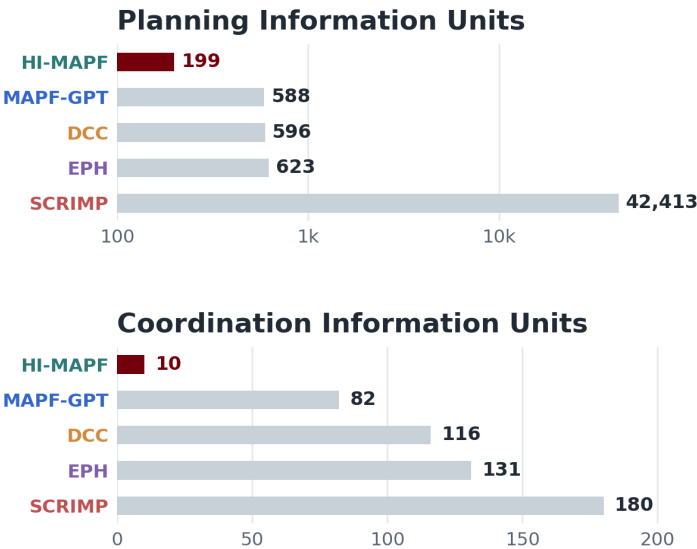
POSTER RADAR



10 messages

8.2× fewer than MAPF-GPT

ACTUAL PLANNING COST



100% success

12.2-step makespan · lowest

Thank you.

HI-MAPF: Towards Resource-Efficient Multi-Agent Deployment

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Paper · Project · Demo



Paper



Project



Demo